FORTISAID LAB 9

Microsoft Sentinel SIEM & SOAR Implementation

Lab Objective

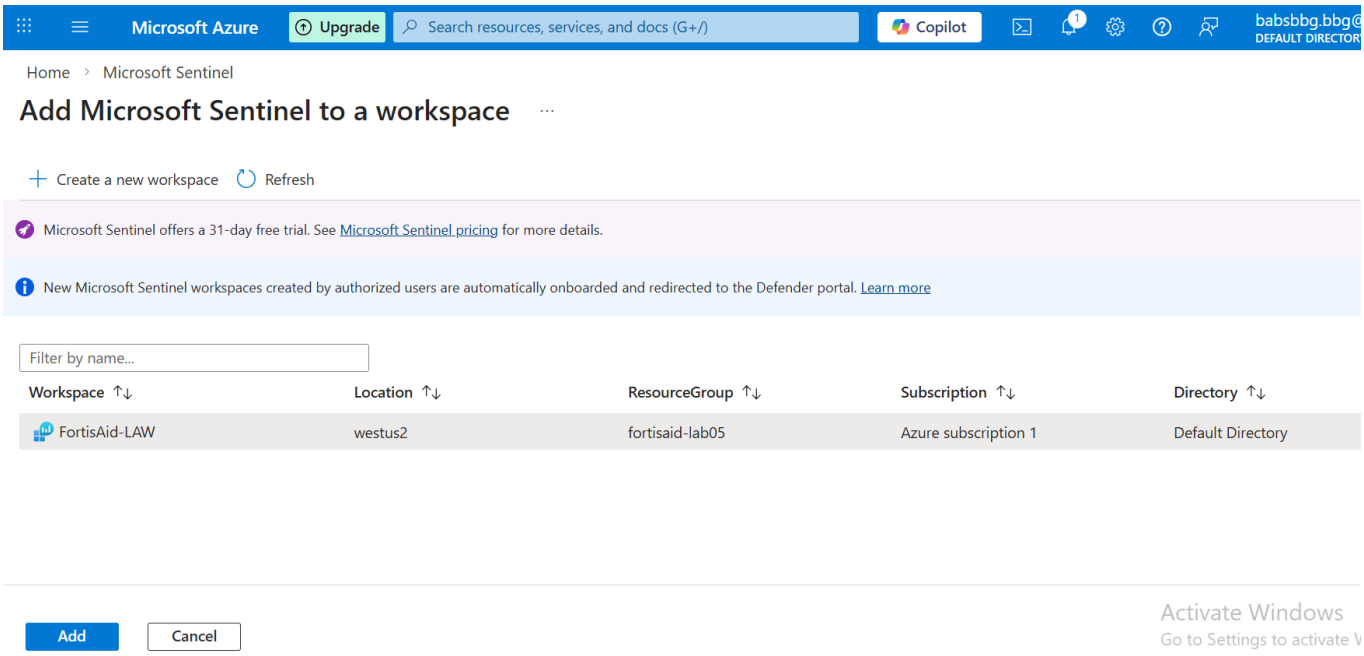
Deploy Microsoft Sentinel on an existing Log Analytics Workspace, ingest Azure Activity logs, create detection rules, and implement a SOAR playbook that automatically escalates incident severity following suspicious administrative activity.

This lab demonstrates an **end-to-end cloud SOC detection and response pipeline**.

Task 1 — Onboard Microsoft Sentinel

Steps

1. Search **Microsoft Sentinel**
2. Click **Create**
3. Select existing LAW in **RG: FortisAid-Lab05**
4. Click **Add**

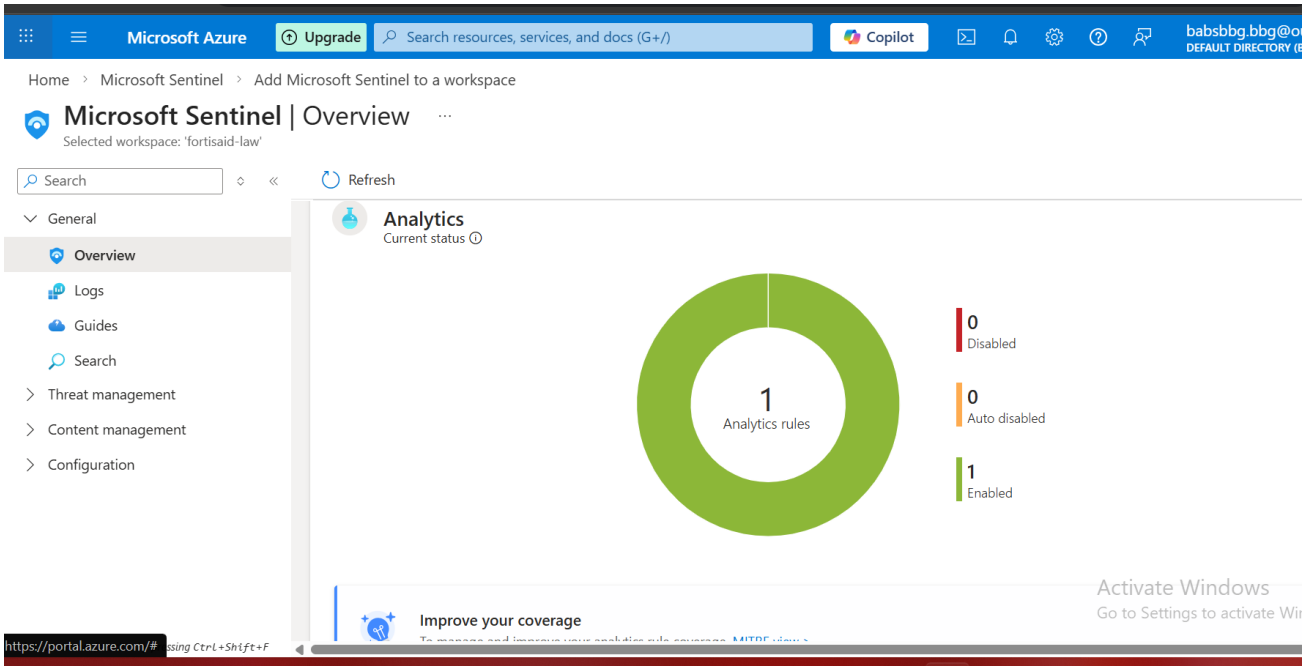


Sentinel Overview showing workspace successfully onboarded

Verification

- Sentinel Overview loads

- Workspace successfully linked

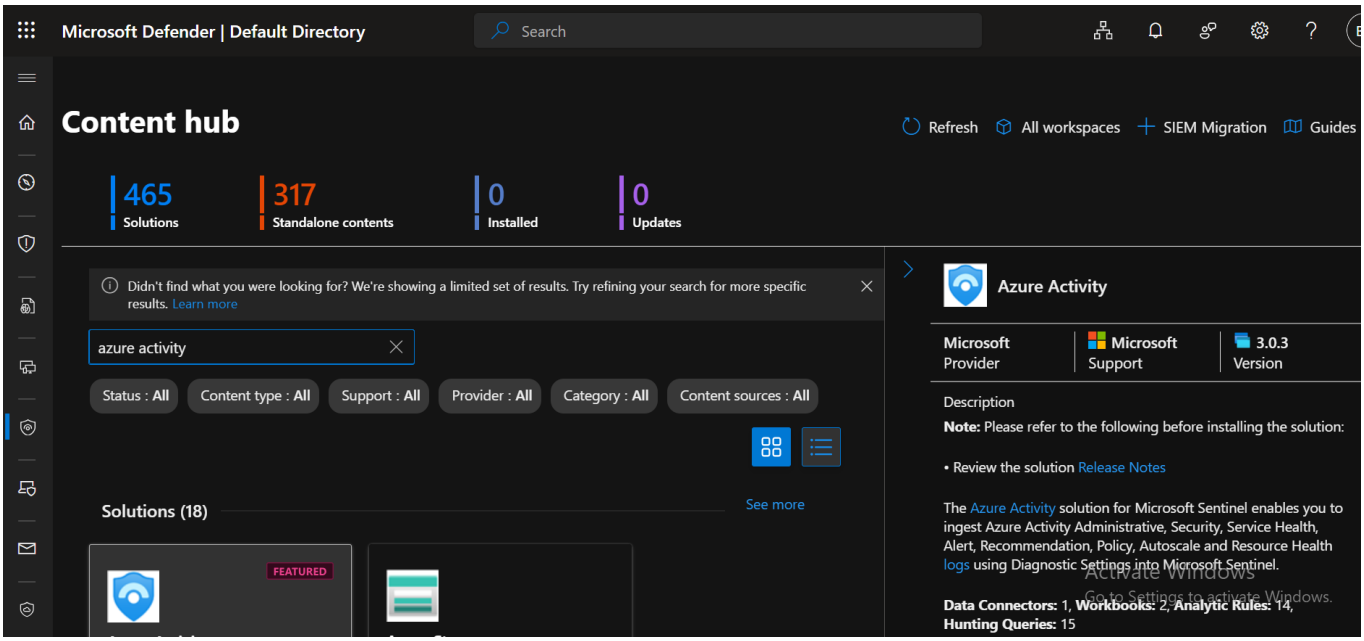


Task 2 — Connect Azure Activity Logs

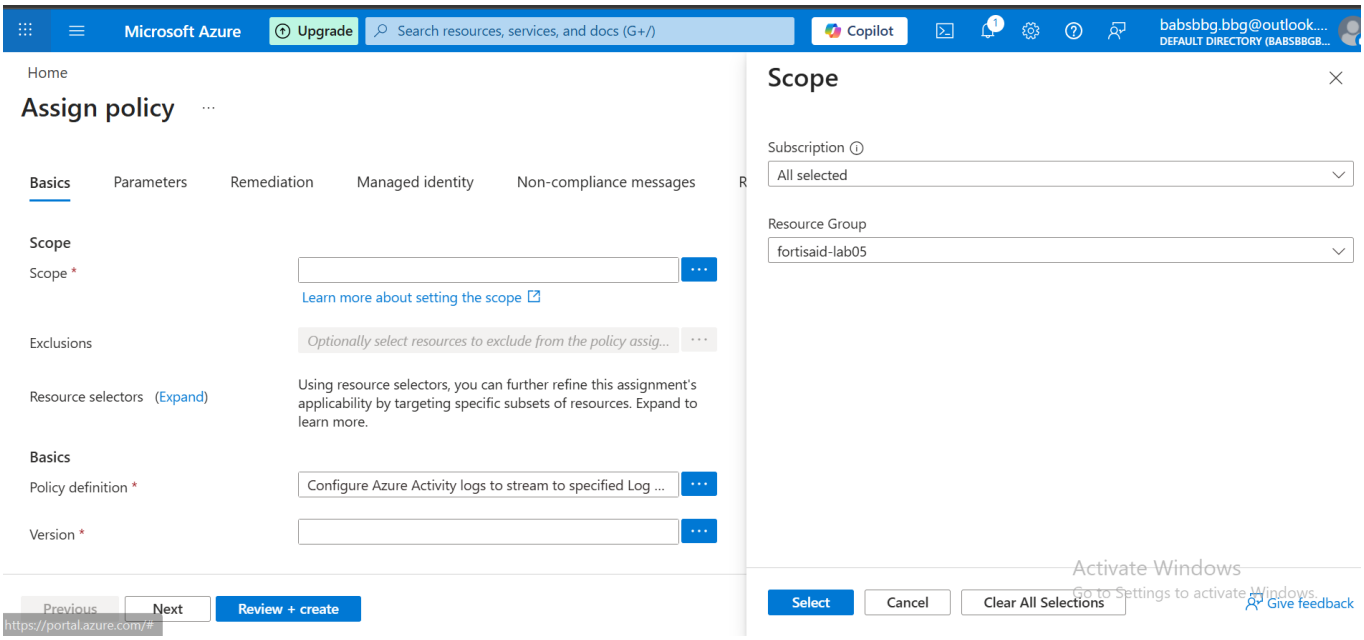
Steps

1. Sentinel → **Content hub**
2. Search **Azure Activity** → Install
3. Sentinel → **Data connectors** → Azure Activity → Open
4. Launch **Azure Policy Assignment wizard**
5. Scope = **Subscription**
6. Parameter = Select LAW
7. Enable remediation task
8. Create policy assignment

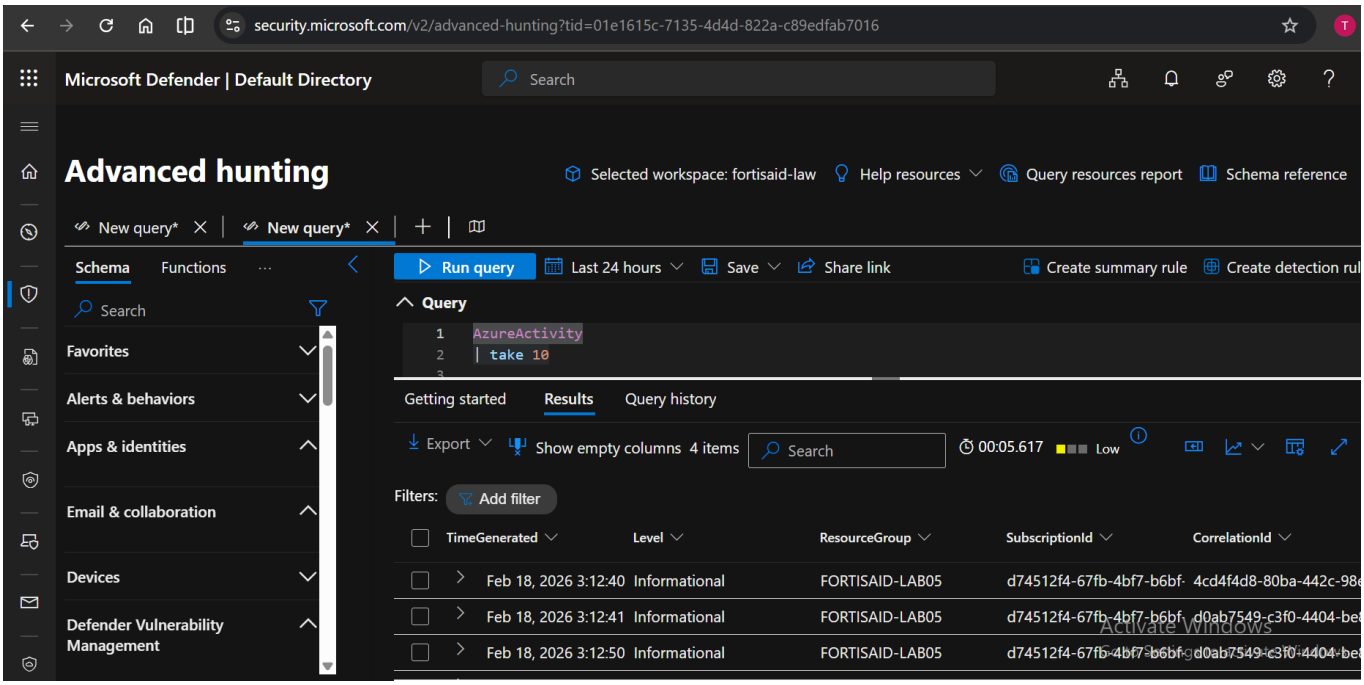
Manual diagnostic setting verification performed due to policy propagation delay.



Azure Activity connector installed



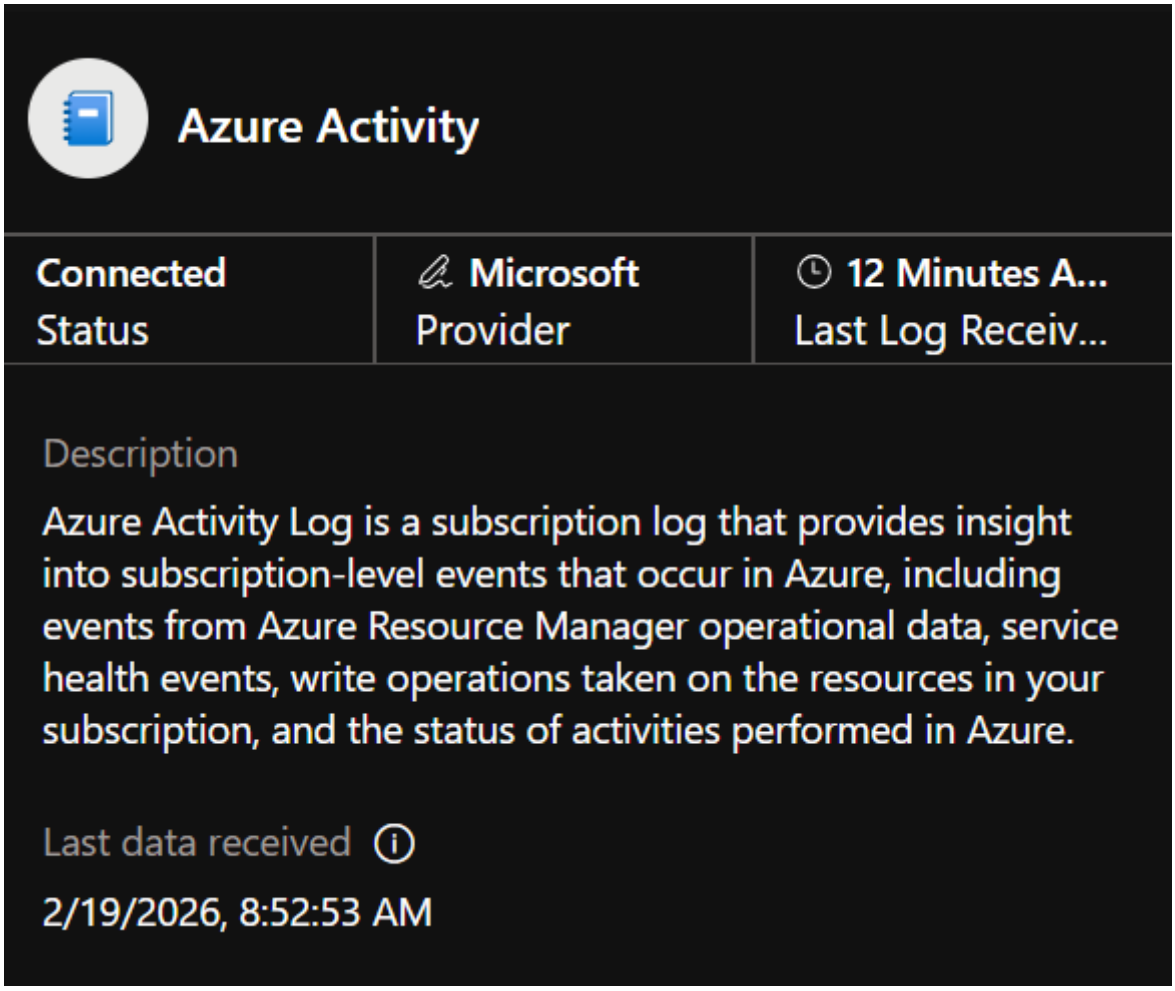
Policy assignment wizard configuration



AzureActivity logs visible in LAW query results

Verification

- AzureActivity table populated

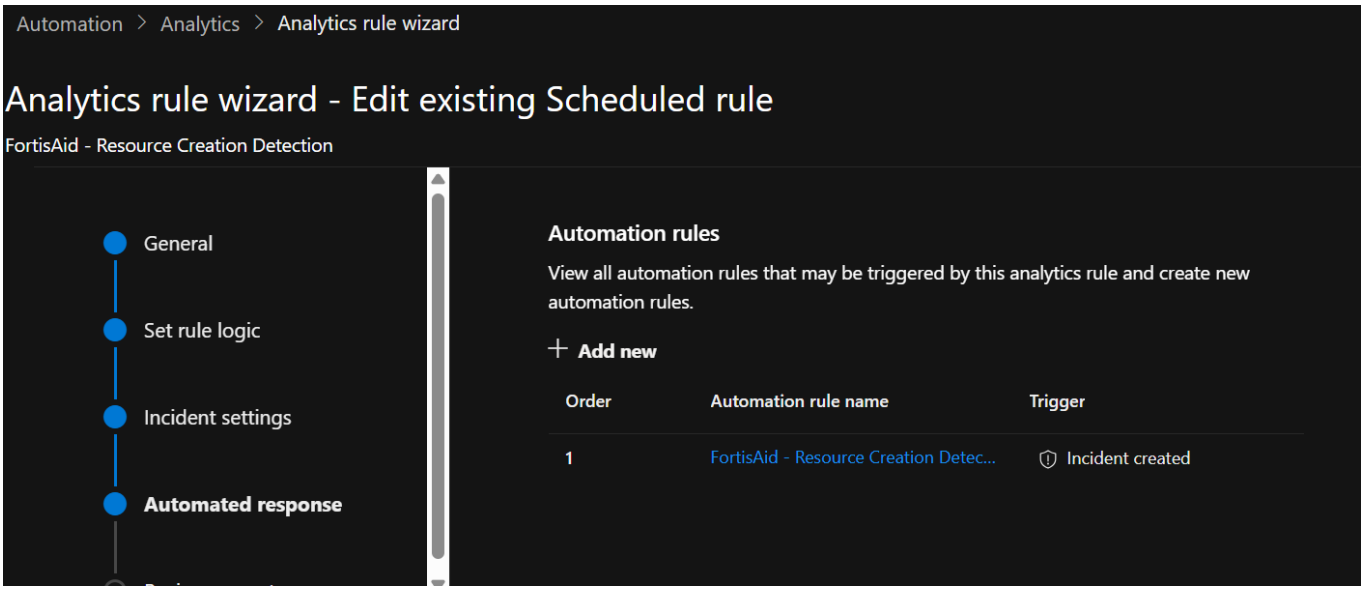


- Connector shows last data received timestamp

Task 3 — Create Detection Rule Using Azure Activity

Steps

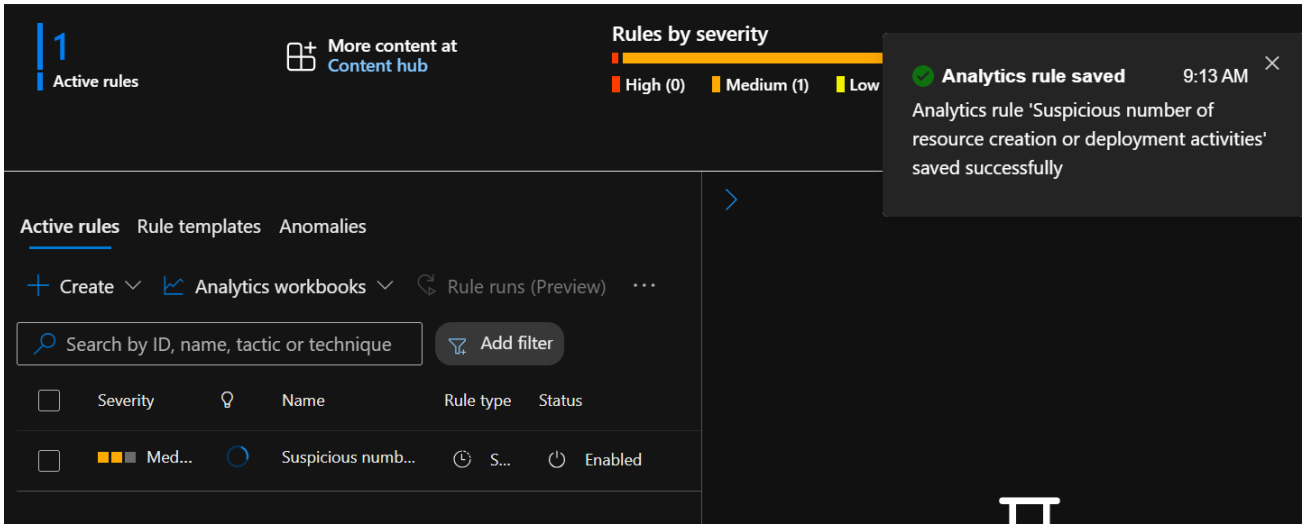
1. Sentinel → **Analytics** → **Rule templates**
2. Deploy **Suspicious number of resource creation or deployment**
3. Save with defaults



Analytics rule active in Sentinel

Verification

- Rule appears in Active Rules

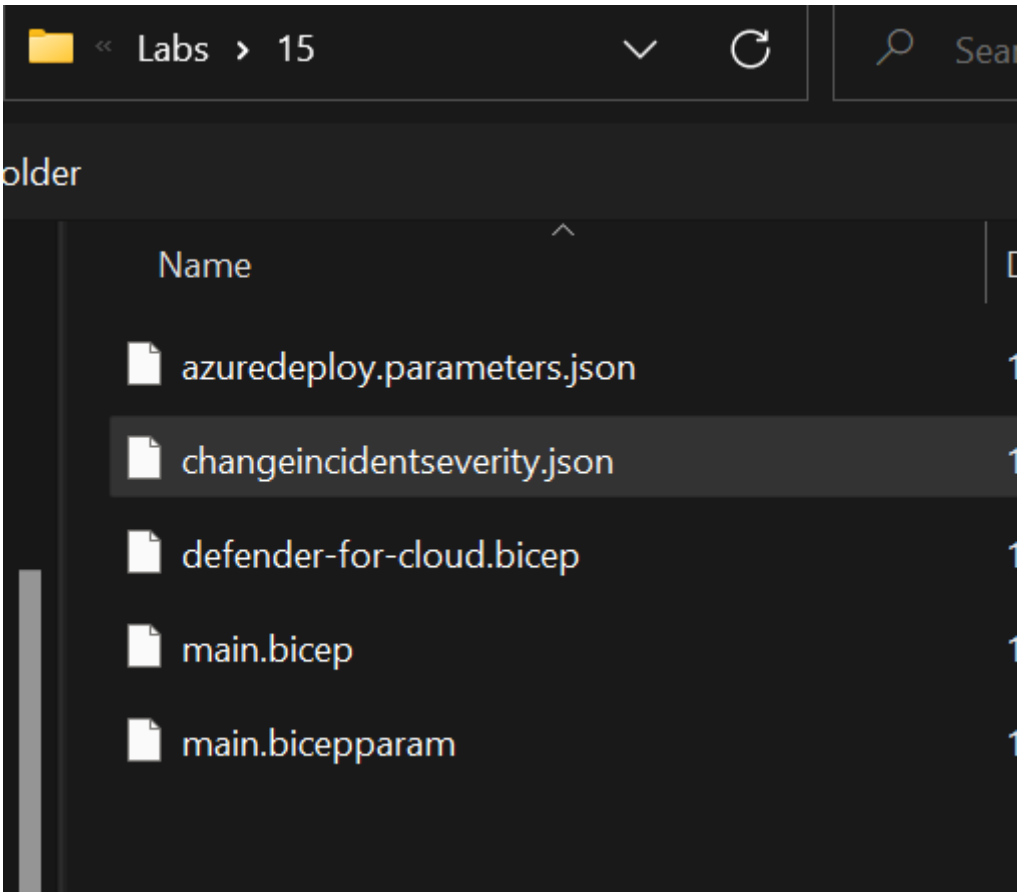


- Incident created during resource creation test

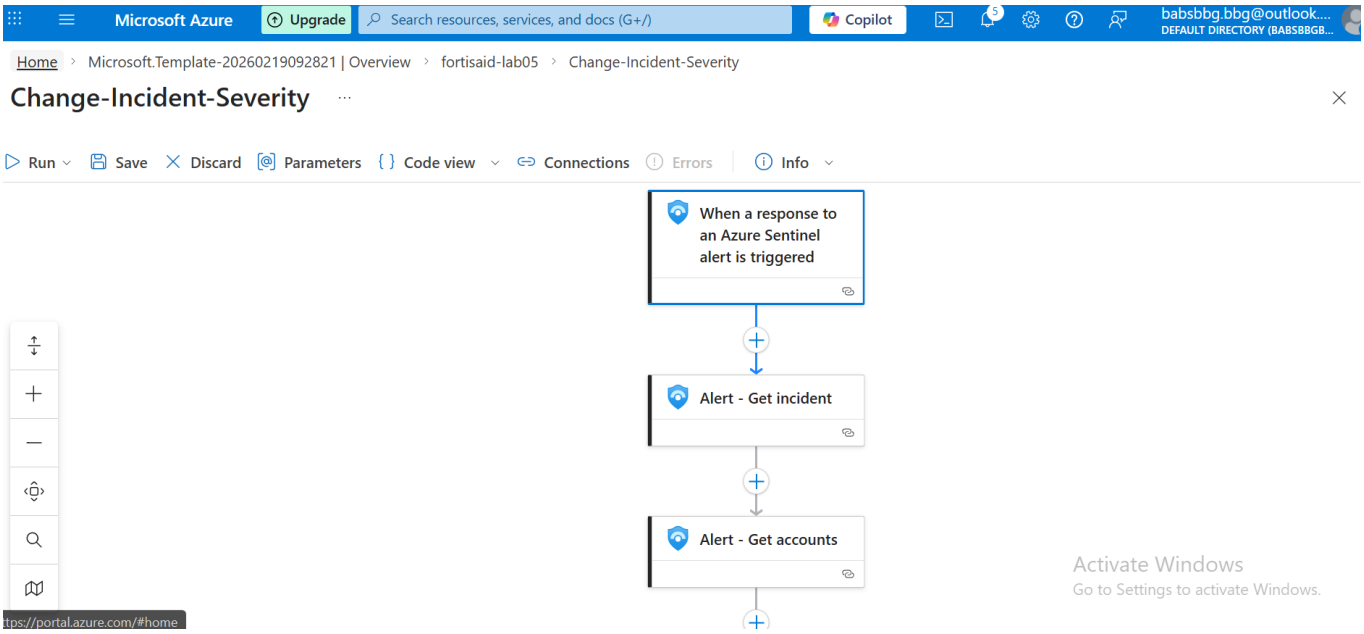
Task 4 — Create SOAR Playbook

Steps

1. Deploy Logic App playbook **Change-Incident-Severity**
2. RG: FortisAid-Lab05
3. Open Logic App → Edit
4. Authenticate Sentinel connectors
5. Add action → **Update Incident**
6. Set severity = High
7. Save



*Logic App playbook deployment



*Logic App designer showing trigger + update incident action

Verification

- Logic App shows 1 trigger + 1 action
- No connection warnings

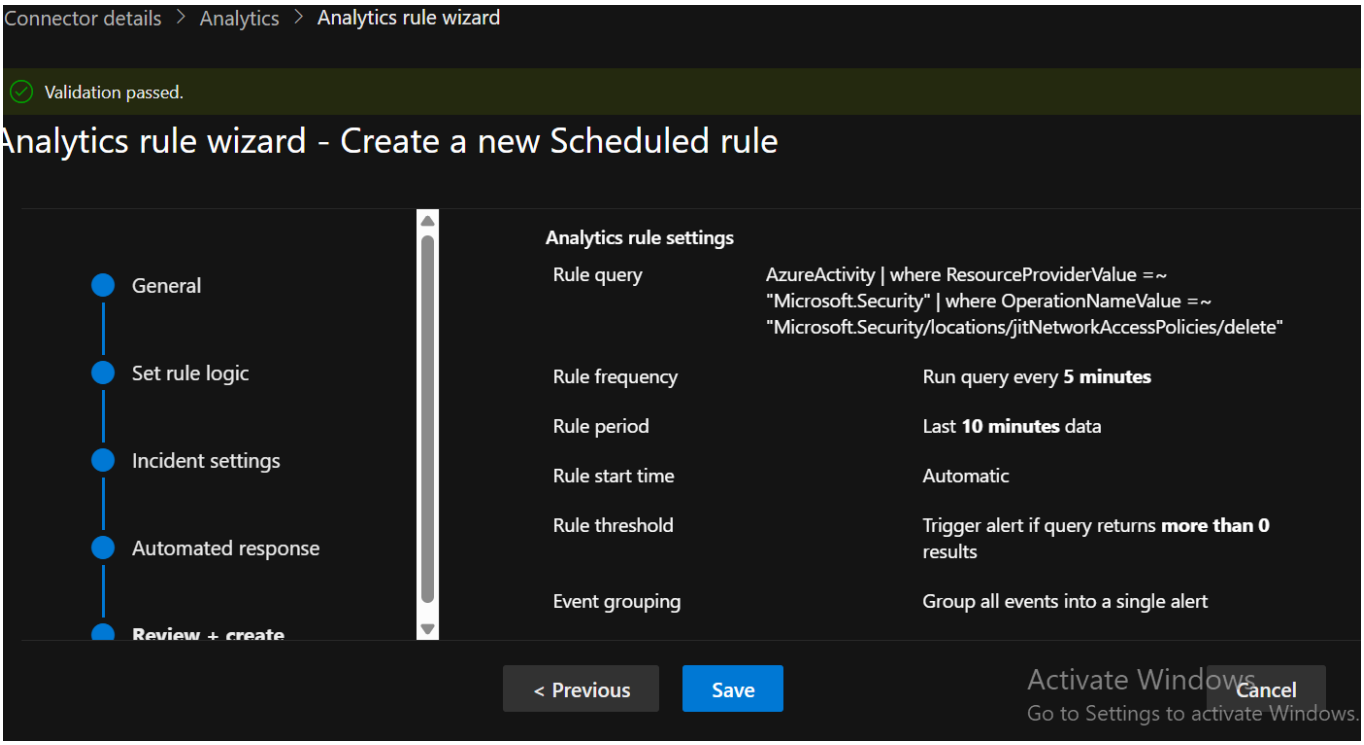
Task 5 — Create Custom Detection + Attach Playbook

Steps

1. Sentinel → Analytics → Scheduled query rule
2. Name: **Playbook Demo**
3. Query:

```
AzureActivity
| where ResourceProviderValue =~ "Microsoft.Security"
| where OperationNameValue =~ "Microsoft.Security/locations/jitNetworkAccessPolicies/delete"
```

4. Frequency = 5 minutes
5. Enable incident creation
6. Attach playbook via automation rule



Custom analytic rule configuration

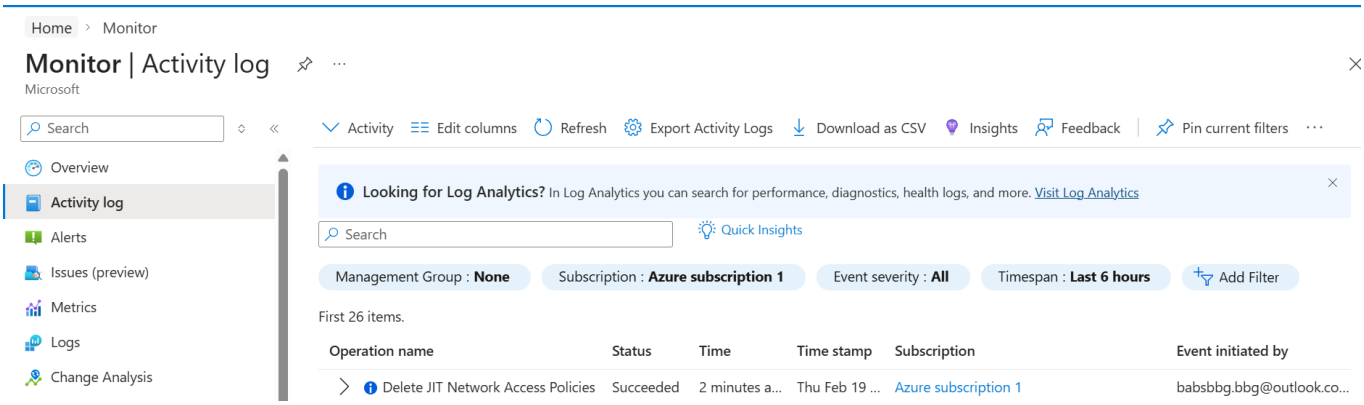
Verification

- Rule enabled
- Automation rule visible
- Permissions configured

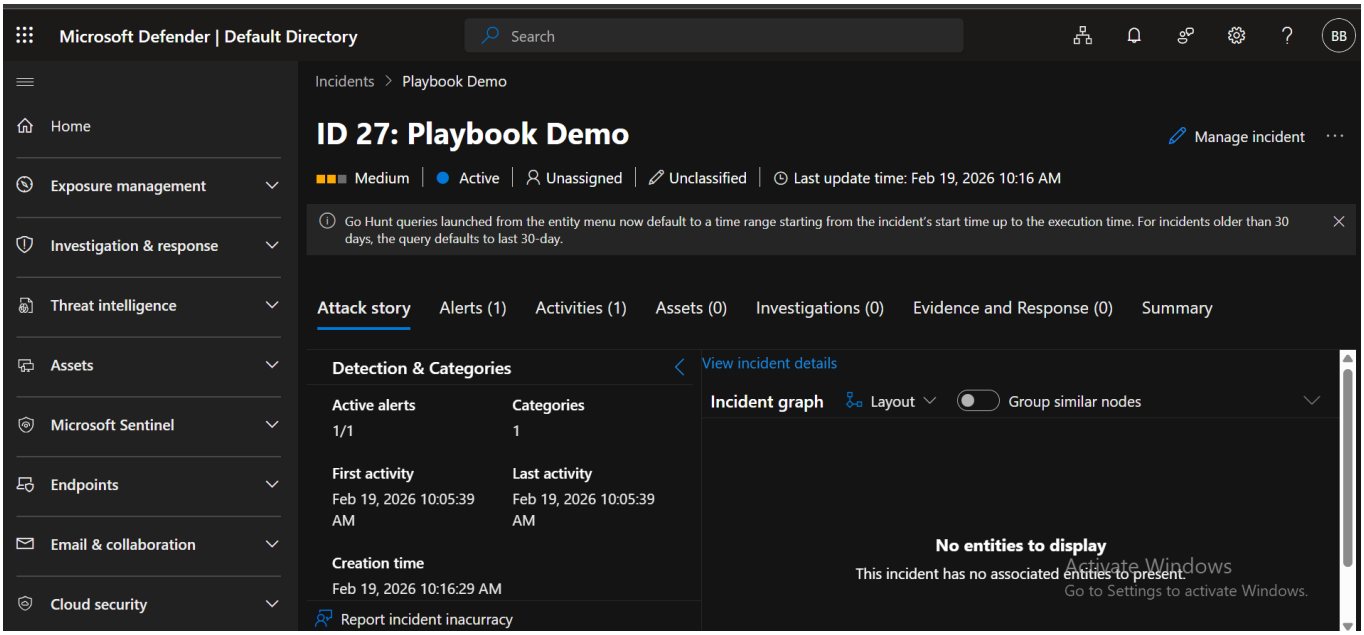
Task 6 — Trigger Incident & Validate SOAR

Steps

1. Defender for Cloud → Remove JIT policy
2. Confirm activity log event
3. Wait for Sentinel rule cycle
4. Open Sentinel → Incidents
5. Confirm playbook run



JIT deletion activity log event

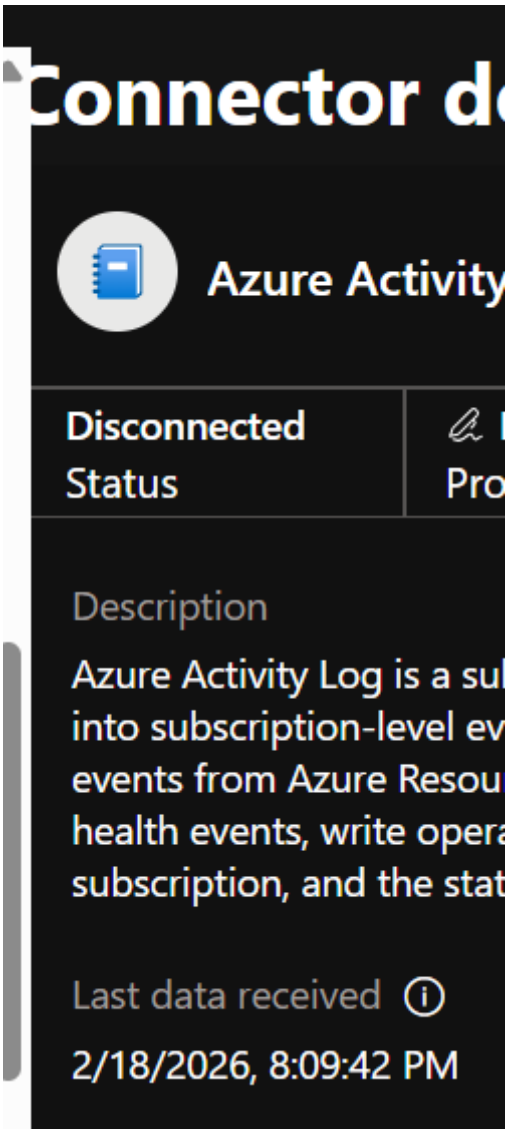


Sentinel incident created

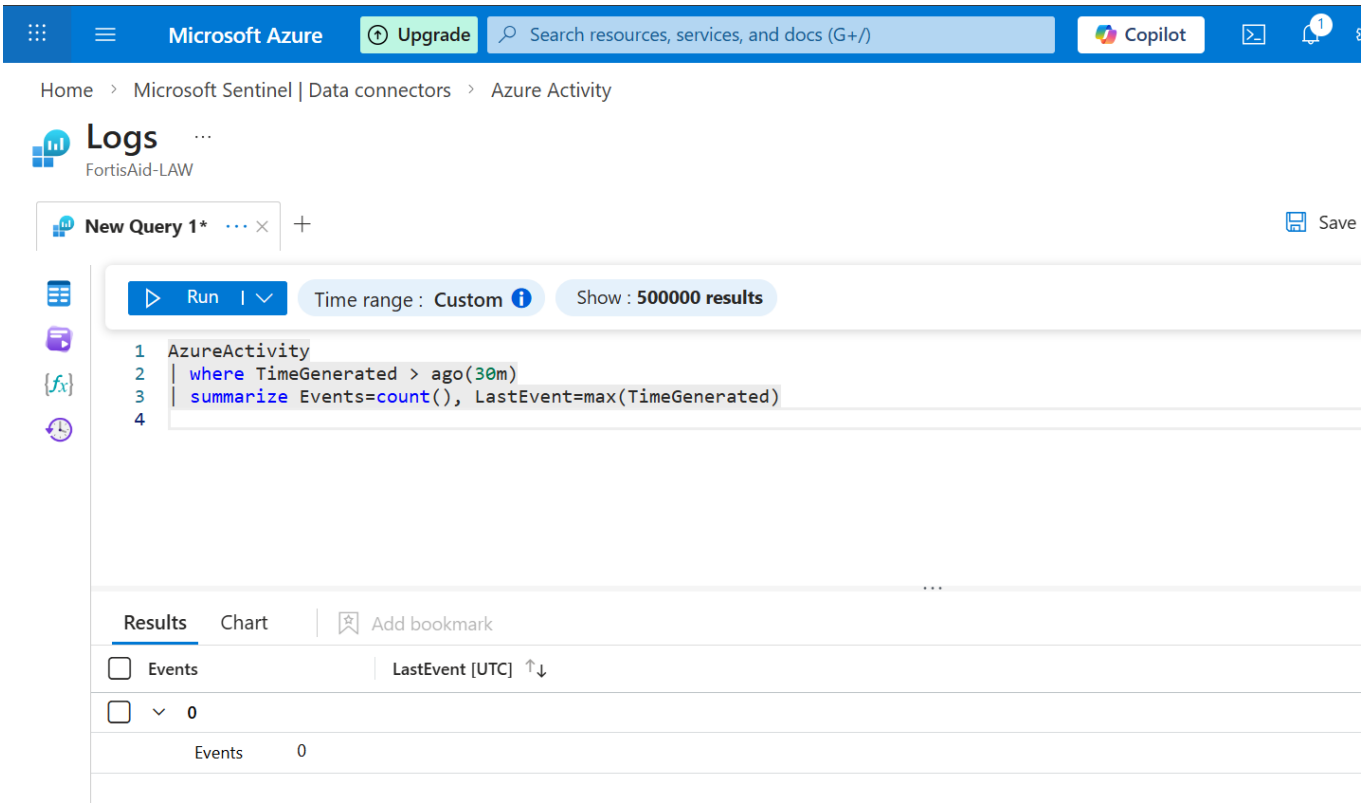
Verification

- Incident created
- Severity escalated automatically
- Logic App run succeeded

Challenges Encountered



Azure Activity connector disconnected state



Analytics rule returning zero results due to ingestion delay

- Azure Policy remediation delay
- Sentinel connector status inconsistency
- Analytics rule ingestion timing mismatch
- RBAC scope blocking playbook execution
- Incident severity not updating until automation configured

Key Technical Learnings

- Azure Activity ingestion delay impacts detection reliability
- Sentinel automation requires subscription-level RBAC
- Incident severity originates from analytic rule
- Playbooks execute only after incident creation
- End-to-end validation requires correlating ingestion → detection → automation

Outcome of the lab

Successfully implemented:

- Sentinel onboarding
- Azure Activity ingestion
- Detection rule deployment
- Incident generation

- SOAR playbook automation
- Automated severity escalation

Demonstrated a complete **cloud SIEM + SOAR detection and response workflow**.